Carbohydrates

● Structurally defined as **aldehydes** or **ketones** with **multiple hydroxyl** groups or "hydrate" of carbon **(CH_2O).**

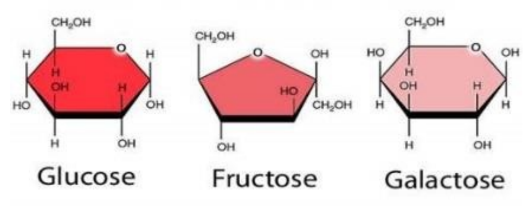
Classification of Carbohydrates;

1- Monosaccharides:

● Are the simplest carbohydrates in that they cannot be hydrolyzed to smaller carbohydrates.

● **Monosaccharides** consist of:
- **3-6 carbon** atoms: (trioses, tetroses, pentoses, and hexoses).
- A **carbonyl group** **aldehyde** (aldose) or **ketone** (ketose).
- Several **hydroxyl groups**.

● **Examples** of ***monosaccharides***: **Glucose**, **Fructose**, **Galactose**.



2- Disaccharides:

● Are formed as a result of combination of **2 monosaccharide** units by **glycosidic bond**.

● **Glycosidic bonds** formed from reaction of monosaccharide with **alcohol** or with **amine** as follows:

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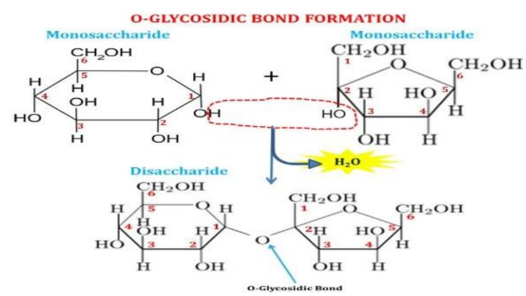
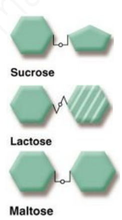
● They can yield two monosaccharides on **hydrolysis**.



● **Examples** of ***disaccharides***:

- **Sucrose** ("table sugar")\\ **glucose** + **fructose**
- **Lactose** ("milk sugar")\\ **glucose** + **galactose**
- **Maltose** ("malt sugar")\\ **glucose** + **glucose**



3- Oligosaccharides:

● Short carbohydrate chains of **3 - 10** monosaccharides.

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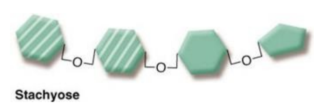
● Found in **legumes** and human **milk**.

● Their function in living organisms is usually either structure or **storage**.



● **Examples** of oligosaccharides: **[Raffinose** and **Stachyose],** cannot be broken down by human enzymes, though can be **digested by colonic bacteria**.

4- Polysaccharides:

● Polysaccharides contain **greater than 10 mono saccharide** units.

● They represent an important class of biological **polymers**.

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- Polysaccharides is hydrolyzed when the demand for sugar increase. hydrolyzed as needed to provide sugar for cells.

● **Examples** of polysaccharides: **Starch**, **Glycogen**, and **Cellulose**.

Differences between polysaccharides:

Polysaccharides	Sources	Structures	Digestions
Glycogen	Animals	Polymers of glucose **(α1-4, and α1-6),** highly branched chain	Digested by human enzymes.
Starch (amylopectin & amylose)	Plants	Polymers of glucose **(α1-4,and α1-6)**	Digested by human enzymes
Cellulose	Plants	Polymers of glucose **(β1-4)**	Can't be digested by human enzymes

● **Glycosaminoglycans**, also known as **mucopolysaccharides**, made of repeating units of **disaccharides** containing a derivative of an amino sugar.

● One of the sugars in unit has **negatively charged** carboxylate or sulfate group.

● Usually attached to proteins such as **proteoglycans**.

Reducing Sugars & Non Reducing Sugars:

● A **reducing sugar** is any sugar that is capable of acting as a reducing agent.

● It has a **free aldehyde** group or a **free ketone** group.

● All monosaccharides are reducing sugars, along with some disaccharides, oligosaccharides, and polysaccharides.

● Examples of a reducing sugar that react with an oxidizing agent: **Glucose** $\xrightarrow{\text{Oxidation reduction reaction}}$ **Gluconic Acid**

Complex & Simple Carbohydrates:

● Complex carbohydrates are often referred to as starch and starchy foods.

● Simple carbohydrates are also known as sugars.

Digestion of Carbohydrates:

● Digestion of carbohydrates begins in the **mouth**.

● **Salivary α- amylase** hydrolyze **α1-4 bonds** of dietary **polysaccharides**, resulting dextrins.

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● Carbohydrate digestion halts temporarily in the stomach, because the high acidity inactivates the salivary α- amylase.

● In the small intestine, the acidic stomach neutralized by **bicarbonate** secreted by the pancrease, and **pancreatic α- amylase** continues the process of starch digestion.

● The final digestive processes occur at the **mucosal lining of the upper jejunum**, which includes the action of several disaccharidases and oligosaccharidases.

- isomaltose $\xrightarrow{\text{isomaltase}}$ glucose
- maltose $\xrightarrow{\text{maltase}}$ glucose
- sucrose $\xrightarrow{\text{sucrase}}$ glucose + fructose
- lactose $\xrightarrow{\text{lactase}}$ galactose + glucose

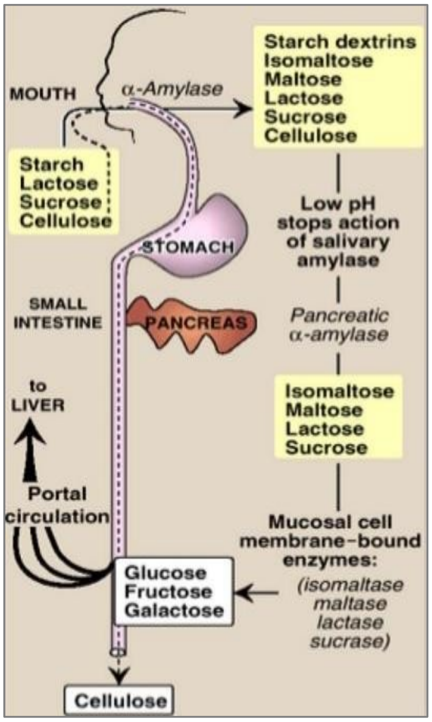
Absorptipon of Carbohydrates:

● **Glucose**, and **galactose** are transported into the mucosal cells by an **active transport system (SGLT-1)**.

● **Fructose** transported by **passive transport system (GLUT-5)**.

● All three **monosaccharides** are transported from the intestinal mucosal cell into the portal circulation by **(GLUT-2)**.

Disturbances in Carbohydrate Resorption:



1. Disaccharidase deficiency syndrome:

A- Saccharase, lactase, maltase;
1. When activity of disaccharidase ↓
2. Hydrolysis of disaccharide ↓
3. Resorption of substrate ↓
4. Concentration of disaccharide in small intestine lumen ↑
5. Osmotic activity of the lumen fluid ↑
6. Diarrhea.

B-
1. Activity of disaccharidase ↓
2. Concentration of disaccharide in small intestine lumen ↑
3. Concentration of disaccharide in large intestine ↑
4. Disaccharide fermentation by bacteria
5. Concentration of lactic acid and fatty acids ↑
6. Stimulation of intestine wall
7. Abdominal cramps, bloating, diarrhea, acidic stools, explosive diarrhea.

2. Lactase deficiency syndrome:

● Causes of lactase deficiency:

▪ Genetic defect (primary)

▪ Secondary to a wide variety of gastrointestinal diseases that damage the mucosa of the small intestine.

▪ Disaccharide **lactose** is the principal carbohydrate in **milk**.

▪ Many persons showing milk intolerance prove to be **lactase – deficient**.

▪ Milk intolerance may not become clinically apparent until adolescence.

3. Monosaccharides malabsorption:

● Small intestine ability to resorb glucose and galactose is **decreased**.

● **Cause**: Specific transport system for galactose and glucose absorption in cells of small intestine is **insufficient**.

● **Results**: Symptoms and signs **similar to** disaccharidase deficiency syndrome.

Diabetes Mellitus

● **Diabetes Mellitus (DM)** is the commonest endocrine disorder, defined as a syndrome characterized by **hyperglycemia** due to an **insulin resistance** and an absolute or relative lack of insulin.

● Diabetes Mellitus (DM) is the leading cause of **adult blindness**, **renal failure**, **heart attacks**, and **strokes**.

Differences between Type I & Type II DM

Diabetes Mellitus Type I (Insulin dependent DM)	Diabetes Mellitus Type II (Insulin non-dependent DM)
1. Age **< 30 years**	**1.** Age **> 40 years**
2. Cause**: auto immune destruction of pancreatic islet cell carcinoma.**	**2.** Cause**: Insulin Resistance**
3. Insulin**: low or absent**	**3.** Insulin**: normal or high**
4. Weight**: normal or underweight**	**4.** Weight**: overweight or obese**
5. Treatment**: Insulin is necessary**	**5.** Treatment**: changing life style, medications.**
6. Epidemiology**: 10-20%**	**6.** Epidemiology**: 80-90%**

Symptoms of DM:

- frequent urination
- feeling very thirsty and drinking a lot
- feeling very hungry
- feeling very fatigued
- blurry vision
- cuts or sores that don't heal properly

Main symptoms and signs of DM and mechanisms of their onset:

● **Hyperglycemia:**

- Relative or absolute deficiency of insulin effect ↓
- Transport of glucose to muscle and fat cells ↓
- Glycemia. ↑
- Insulin effect ↓
- Gluconeogenesis in liver ↑
- Blood level of glucose ↑
- Glycogenolysis ↑

● **Polyuria:**

▪ High blood level of glucose
▪ ↑ Amount of glucose filtered by the glomeruli of the kidney
▪ ↑ Absorption capacity of renal tubules for glucose
▪ Glycosuria results, accompanied by large amounts of water lost in the urine (osmotic effect of glucose).

● **Polydipsia:**

▪ High blood level of glucose
▪ Hyperosmolality of plasma
▪ Water moves from cells to ECF
▪ Intracellular dehydratation
▪ Creation of thirst feeling (in hypothalamus)
▪ *Intake of fluids.*

● **Polyphagia:**

▪ Depletion of cellular stores of carbohydrates, fats, and proteins →
▪ Cellular starvation and a corresponding increase in hunger.

● **Weight loss:**

▪ Fluid loss in osmotic diuresis, loss of body tissue as fats and proteins are used for energy creation.

● **Fatigue:**

▪ Metabolic changes result in poor use of food products
▪ Lethargy and fatigue

Important biochemical changes that occur in uncontrolled DM:

▪ Increased blood glucose level
▪ Hyper cholesterolaemia
▪ Glucoseuria
▪ Ketonaemia
▪ Incresed urea and non protein nitrogen compound
▪ Dehydration
▪ Acidosis (decreased PH and hyperventilation) and kussmaul breathing.
▪ Lowered sodium and disturbance in fluid and electrolyte balance.

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